

Review copy: September 7, 2026. This document revises the existing Phase 1 module. It is not the final manuscript. User approval is pending.

1 Primitives, Objects, and Timing

This section fixes the objects and information structure of the baseline before any payoff is constructed. The model is a partial equilibrium of viable drug-project advancement, commercialization organization, and qualified manufacturing services. The MAH reform changes only the institutional feasibility or friction of retained entrusted manufacturing.

1.1 Units and populations

Let C denote currency units, X project-advancement input units, P the expected measure of viable planning-stage projects in one decision cohort, K manufacturing capability/requirement units, and B qualified manufacturing-service capacity units.

There is a continuum $\mathcal{I}$ of drug developers. Developer i has research capability $a_i > 0$, measured in P/X , and internal manufacturing capability $k_i > 0$, measured in K . Its characteristics follow the exogenous distribution $H(a, k)$. Its complete permanent baseline characteristic vector is

$$\theta_i = (a_i, k_i). \quad (1)$$

These are continuous characteristics, not permanent discrete firm types.

There is also a set $\mathcal{J}$ of qualified manufacturing-service suppliers. Supplier efficiency $z_j > 0$ follows the exogenous distribution $H_C(z)$, where z is a generic supplier-efficiency value. The supplier's capacity control and cost function are introduced when the qualified manufacturing-service market is defined below.

1.2 Project characteristics and empirical classification

A viable project that reaches the route-planning stage draws commercial or scientific value shifter $q > 0$ and manufacturing requirement $m > 0$. The latter is measured on the same K scale as k_i . Let F denote the exogenous joint distribution of project characteristics. Define

$$\omega = (q, m) \sim F(q, m), \quad (2)$$

where F is an exogenous joint distribution. Thus q governs the project's value side, while the pair (m, k_i) governs the manufacturing mismatch. The institutional regime does not directly alter q , m , or F .

For empirical decompositions only, let $g \in \{O, \text{Inc}\}$ be an exogenous novelty classifier. Here O denotes original/breakthrough-oriented projects and Inc denotes incremental/improved projects. The latter label is distinct from the internal-manufacturing route label defined below. Let $\rho_g \geq 0$ be the class share and $F_g(q, m)$ the conditional project distribution. They satisfy

$$\sum_{g \in \{O, \text{Inc}\}} \rho_g = 1, \quad F(q, m) = \sum_{g \in \{O, \text{Inc}\}} \rho_g F_g(q, m). \quad (3)$$

The classifier is not a new baseline state, and it does not create a separate advancement control for each class.

1.3 Institutional regime and route labels

Let the exogenous institutional regime be $M \in \{0, 1\}$. The value $M = 0$ denotes the pre-MAH regime. The value $M = 1$ means that a developer may legally retain authorization while a qualified external producer manufactures the product. Let $\tau_E(M)$, measured in C per project, denote the institutional barrier attached only to this retained entrusted route. The finite post-MAH burden is denoted by $\bar{\tau}_E \geq 0$. The baseline representation is

$$\tau_E(0) = +\infty, \quad \tau_E(1) = \bar{\tau}_E < +\infty, \quad \bar{\tau}_E \geq 0. \quad (4)$$

There is no continuous implementation parameter in the baseline.

Let p_m , measured in C/B, denote a conjectured price of qualified manufacturing service and let p_m^* denote its future market-consistent value. Let $\Omega_i(M, p_m)$, measured in C per planning-stage project, denote developer i 's expected optimized project value. These objects are semantically reserved here: the sections below first build route values, then solve the advancement problem, and finally close the CMO fixed point.

The route labels are I for internal manufacturing, E for retained entrusted manufacturing, T for transfer or another non-retained outside option, and A for abandonment or indefinite delay. If r_i denotes the project's route control, its common label domain is

$$r_i \in \{I, E, T, A\}. \quad (5)$$

Effective pre-MAH unavailability of E is encoded solely by $\tau_E(0) = +\infty$, which makes the entrusted route unavailable when values are assembled. Thus the common label domain is not a second policy channel. Under E , the developer remains the holder; entrusted production is not a transfer of ownership or authorization. The eventual optimizer is denoted by $r_i^*(q, m; M, p_m)$, but route values and the argmax are defined in the organizational-sorting section below.

1.4 Project-advancement and realization objects

Developer i chooses one common control $x_i \geq 0$, measured in X. Canonically, x_i is *project-development / advancement intensity*: effort that advances viable drug projects to the stage at which commercialization routes are planned. Upstream scientific research, basic-compound discovery, patent-generating effort, and patent applications are outside this control. Let x_i^* reserve the optimal value of this common control; its optimization is derived in the project-advancement section.

Research capability converts the control into the expected measure of viable planning-stage projects:

$$\lambda_i^{\text{plan}} = a_i x_i. \quad (6)$$

The control x_i has units X, so λ_i^{plan} has units P.

Finally, let $s(q) \in [0, 1]$ be the exogenous, route-independent probability of downstream clinical or regulatory realization:

$$\Pr\{\text{downstream realization} \mid q\} = s(q). \quad (7)$$

The reform does not shift $s(q)$. If typed observed outcomes later require a class-specific probability $s_g(q)$, it is likewise exogenous and MAH-invariant.

1.5 Baseline timing

- Stage 0. **Institutional regime.** The value of M is realized and publicly observed. It changes only the institutional barrier in (4); the infinite/finite wedge encodes the effective unavailability/availability of route E .
- Stage 1. **Ex ante project advancement.** Developer i observes (a_i, k_i, M) , anticipates the market-consistent service price p_m^* and future route opportunities, and chooses x_i . Equation (6) determines the expected measure of viable projects reaching the route-planning stage.
- Stage 2. **Project characteristics.** Each planning-stage project draws (q, m) according to (2). The optional label g records an empirical class but does not create another advancement choice.
- Stage 3. **Commercialization organization.** After observing (q, m) and the manufacturing-service price, the project chooses a route from the domain in (5). Actual holder–producer separation is observed only if the realized route is E , which occurs after the Stage 1 choice.
- Stage 4. **Downstream realization.** A retained project realizes downstream commercial value with the exogenous probability in (7). Planning-stage projects, route assignments, and realized products are distinct objects.
- Stage 5. **Manufacturing-service consistency.** Aggregate entrusted-capacity demand and qualified-capacity supply must support the same p_m^* that developers anticipated. This is a simultaneous equilibrium consistency condition, not an unanticipated event occurring after Stage 4.

Upstream scientific research and patent generation may have occurred before Stage 0 and may help explain predetermined a_i . They are outside the baseline decision timing.

1.6 Mechanism and exclusion boundaries

The only direct reform arrow is

$$M \longrightarrow \tau_E(M) \quad \text{whose infinite/finite value encodes route } E \text{ availability.} \quad (8)$$

The intended ex ante channel, conditional on the value and optimization results derived below, is

$$M \longrightarrow \text{anticipated availability/value of } E \longrightarrow \Omega_i \longrightarrow x_i^*. \quad (9)$$

The later realization order is

$$\begin{aligned} x_i^* &\longrightarrow \text{planning-stage projects} \longrightarrow r_i^* \\ &\longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.} \end{aligned} \quad (10)$$

Equations (9) and (10) are timing and dependency statements, not unconditional comparative-static propositions.

In particular, M does not directly change a_i , k_i , z_j , q , m , F , ρ_g , F_g , $s(q)$, or $s_g(q)$. It does not directly lower p_m^* . Nothing in this section implies an increase in patent applications, upstream scientific research, or one novelty class relative to the other.

The exact-category registry, dimensions, and forbidden interpretations are fixed by the definitions above. The equilibrium section below closes the deferred fixed point without adding another policy channel.